

2 Subject content

Qualification aims and objectives

The aims and objectives of this qualification are to enable students to:

- understand and apply the fundamental principles and concepts of computer science, including abstraction, decomposition, logic, algorithms, and data representation
- analyse problems in computational terms through practical experience of solving such problems, including designing, writing and debugging programs
- think creatively, innovatively, analytically, logically and critically
- understand the components that make up digital systems and how they communicate with one another and with other systems
- understand the impact of digital technology on wider society, including issues of privacy and cybersecurity
- apply mathematical skills relevant to computer science.

Paper 1: Principles of Computer Science

Overview

In order to become proficient computer scientists, it is essential that students have knowledge and understanding of the field's fundamental principles and concepts.

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